Experiment Number: 1

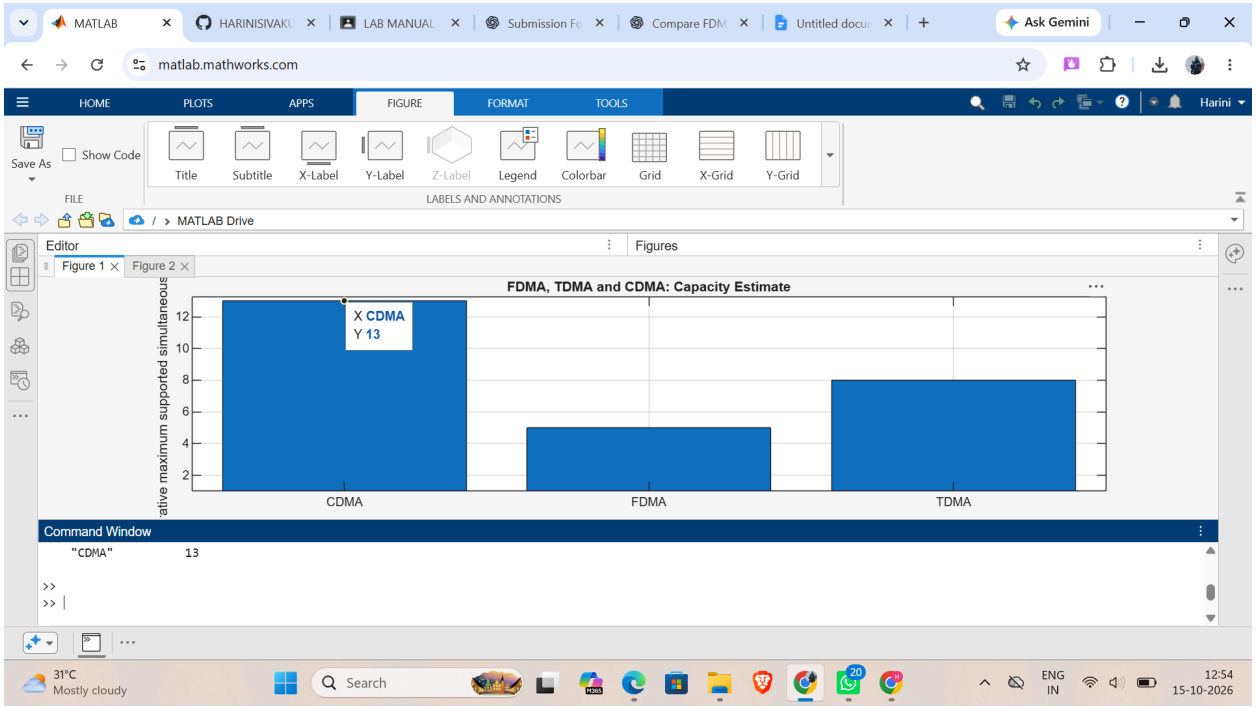
Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context
Program 1 — Capacity estimate for FDMA, TDMA and CDMA
Objective 1: To estimate and compare the illustrative number of users supported by FDMA,
TDMA and CDMA using the specified system parameters.
CODE:

```
clear; clc; close all;
% Illustrative system parameters
totalBandwidthHz = 1e6; % Total bandwidth: 1 MHz
fdmaChannelHz = 200e3; % FDMA channel width: 200 kHz
tdmaSlots = 8; % Number of TDMA slots per frame
processingGain = 64; % CDMA processing gain
targetSIRdB = 7; % Minimum acceptable CDMA SIR
% Capacity estimates under the stated assumptions
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
cdmaUsers, targetSIRdB);
% CDMA SIR as the number of simultaneous users increases
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
% Display a comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];

results = table(method, supportedUsers);
disp(results);
% Plot estimated user capacity
figure;
bar(categorical(method), supportedUsers);
```

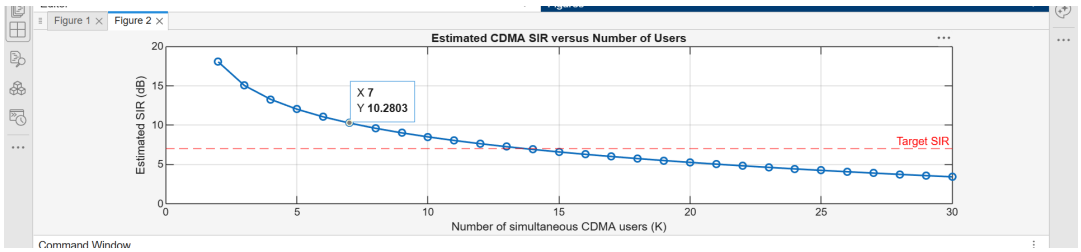
```
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
% Plot estimated CDMA SIR as users are added
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

OUTPUT:



Objective 2: CDMA SIR analysis:

To analyze how the estimated CDMA signal-to-interference ratio changes as the number of simultaneous users increases.



Objective 3: Mobile-generation comparison:

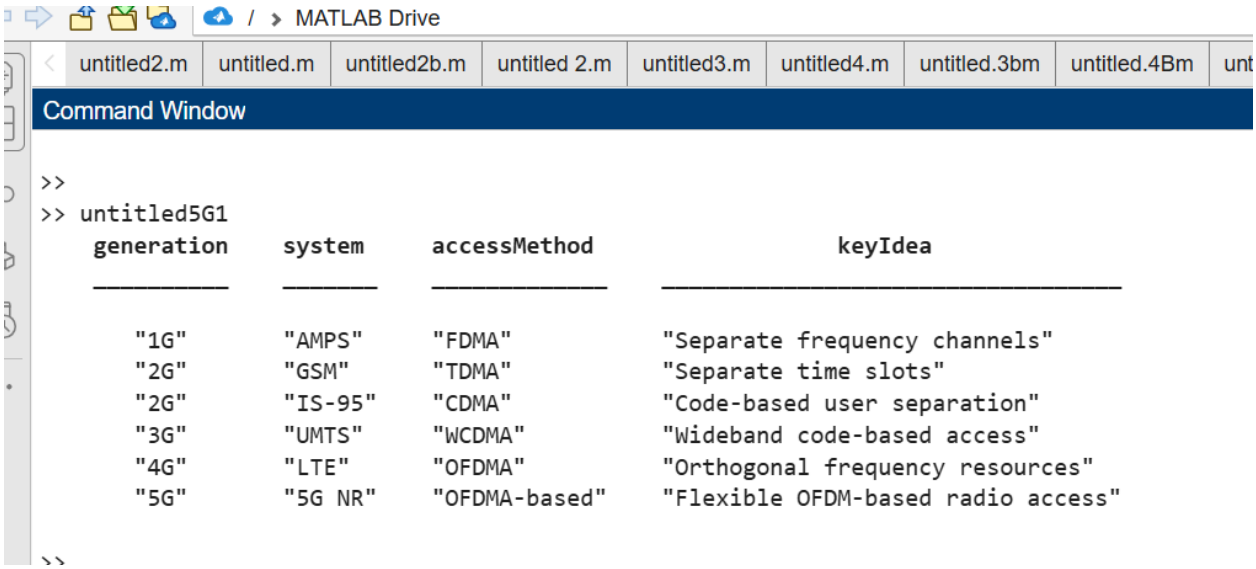
To relate representative multiple-access techniques to the evolution of mobile communication systems from 1G to 5G.

CODE:

```
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];
accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];
keyIdea = ["Separate frequency channels"; ...
"Separate time slots"; ...
"Code-based user separation"; ...
"Wideband code-based access"; ...
"Orthogonal frequency resources"; ...
"Flexible OFDM-based radio access"];

evolutionTable = table(generation, system, accessMethod, keyIdea);
disp(evolutionTable);
```

OUTPUT:



Objective 4: Resource-allocation diagrams:

To visualize how FDMA, TDMA and CDMA assign frequency, time and code resources to
multiple users.

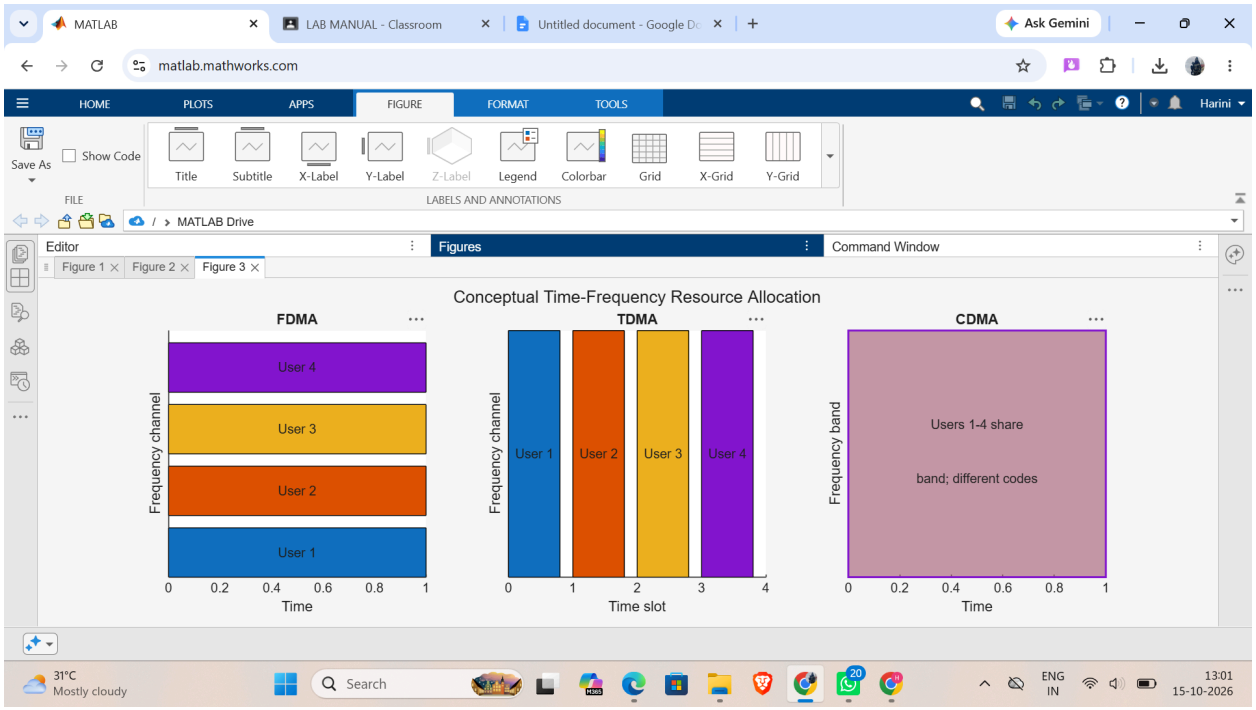
CODE:

```
figure;
tiledlayout(1,3);
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
rectangle('Position',[0, u-1, 1, 0.8], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(0.5, u-0.6, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');

set(gca,'YTick',[]);
hold off;
% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
rectangle('Position',[u-1, 0, 0.8, 1], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(u-0.6, 0.5, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
% CDMA: users share the same time and frequency band using different codes
nexttile;
```

```
hold on;
for u = 1:4
rectangle('Position',[0, 0, 1, 1], ...
'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
xlim([0 1]); ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca,'YTick',[]);
hold off;
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

OUTPUT:



Experiment Number: 2

MATLAB Simulation of Cellular Coverage, Cell Splitting and Handoff in a Multi-Cell

Network

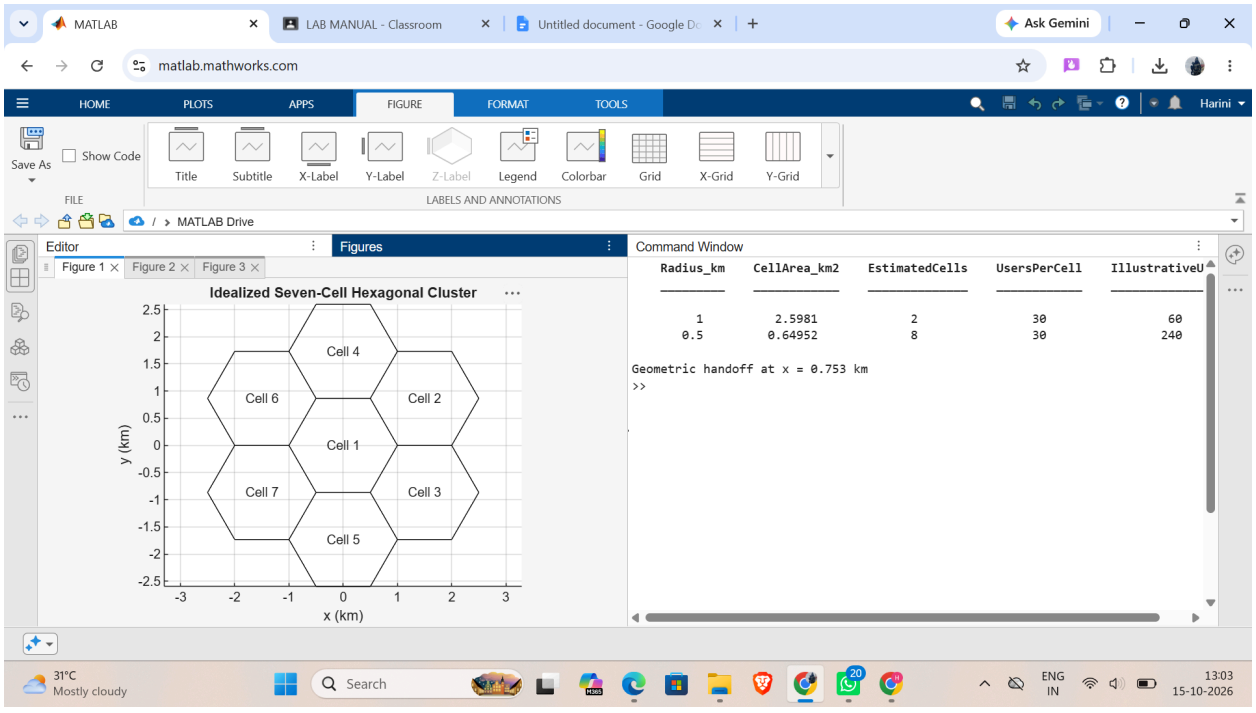
CODE:

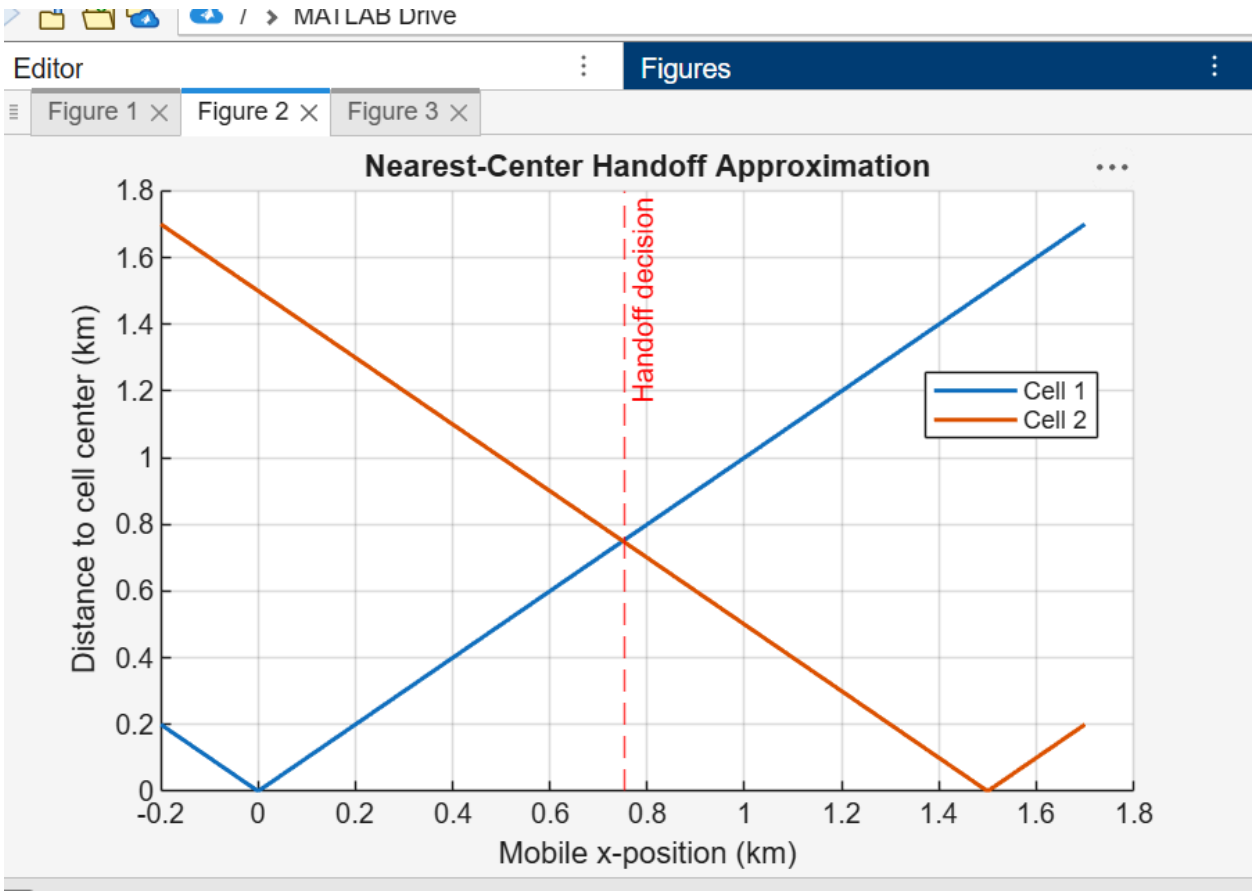
```
% Experiment 2: simplified hexagonal coverage and handoff
clear; clc; close all;
% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
repmat(usersPerCell,numel(radiiKm),1), estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
% Plot a 7-cell cluster: central cell plus six neighbors
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; 1.5*R -sqrt(3)*R/2; ...
0 sqrt(3)*R; 0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
-1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3; % flat-top hexagon vertices
figure; hold on; axis equal; grid on;
for k = 1:size(centers,1)
plot(centers(k,1)+R*cos(theta), centers(k,2)+R*sin(theta), 'k-');
text(centers(k,1),centers(k,2),sprintf('Cell %d',k), ...
'HorizontalAlignment','center');
end
xlabel('x (km)'); ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');

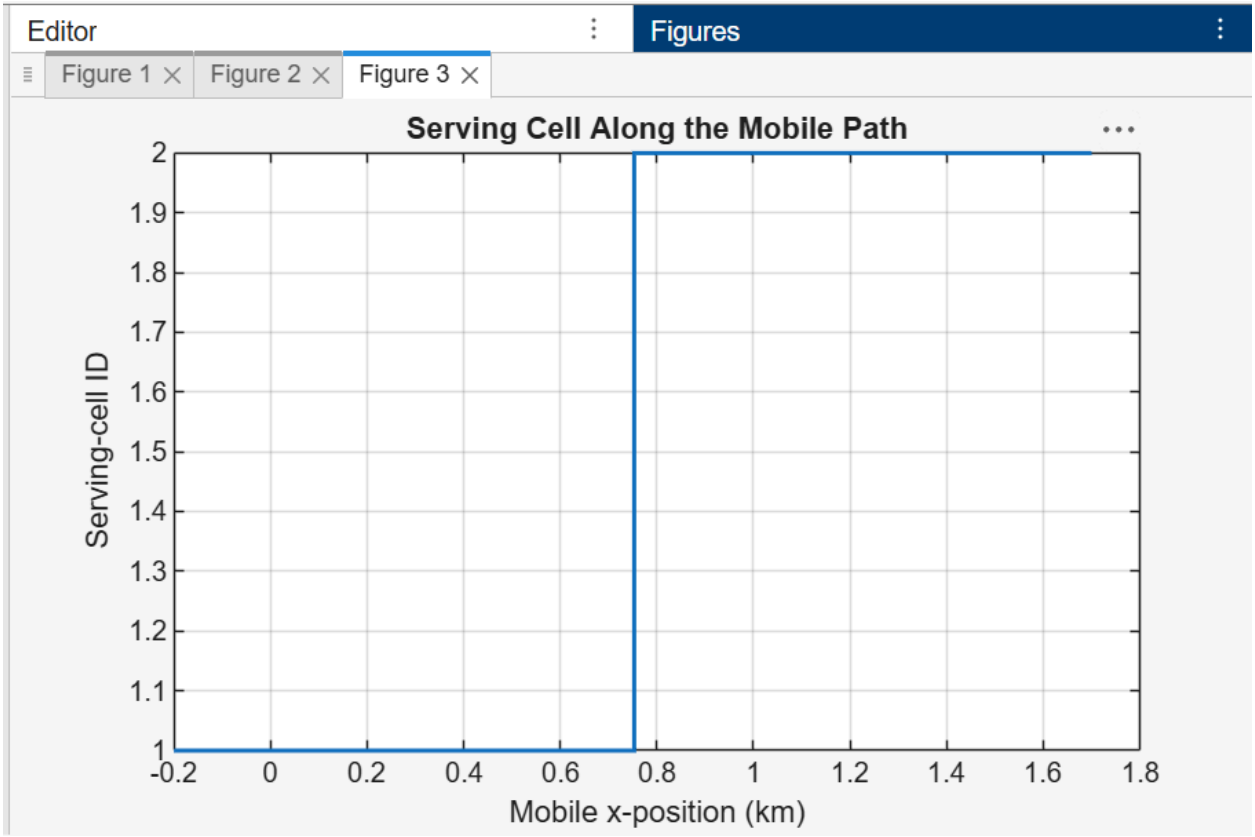
hold off;
% Mobile moves across the boundary between centers at x=0 and x=1.5R
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX-1.5*R, mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
```

```
handoffIndex = find(diff(servingCell) ~= 0,1,'first') + 1;
figure; hold on; grid on;
plot(mobileX,d1,'LineWidth',1.5);
plot(mobileX,d2,'LineWidth',1.5);
if ~isempty(handoffIndex)
xline(mobileX(handoffIndex),'--r','Handoff decision');
end
xlabel('Mobile x-position (km)'); ylabel('Distance to cell center (km)');
legend('Cell 1','Cell 2','Location','best');
title('Nearest-Center Handoff Approximation');
hold off;
figure; stairs(mobileX,servingCell,'LineWidth',1.5); grid on;
xlabel('Mobile x-position (km)'); ylabel('Serving-cell ID');
title('Serving Cell Along the Mobile Path');
if ~isempty(handoffIndex)
fprintf('Geometric handoff at x = %.3f km\n',mobileX(handoffIndex));
end
```

OUTPUT:




MATLAB Drive
Editor
Figures
Figure 1
Figure 2
Figure 3
Nearest-Center Handoff Approximation
Handoff decision
Cell 1
Cell 2
Distance to cell center (km)
Mobile x-position (km)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
-0.2



Experiment Number: 3

MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

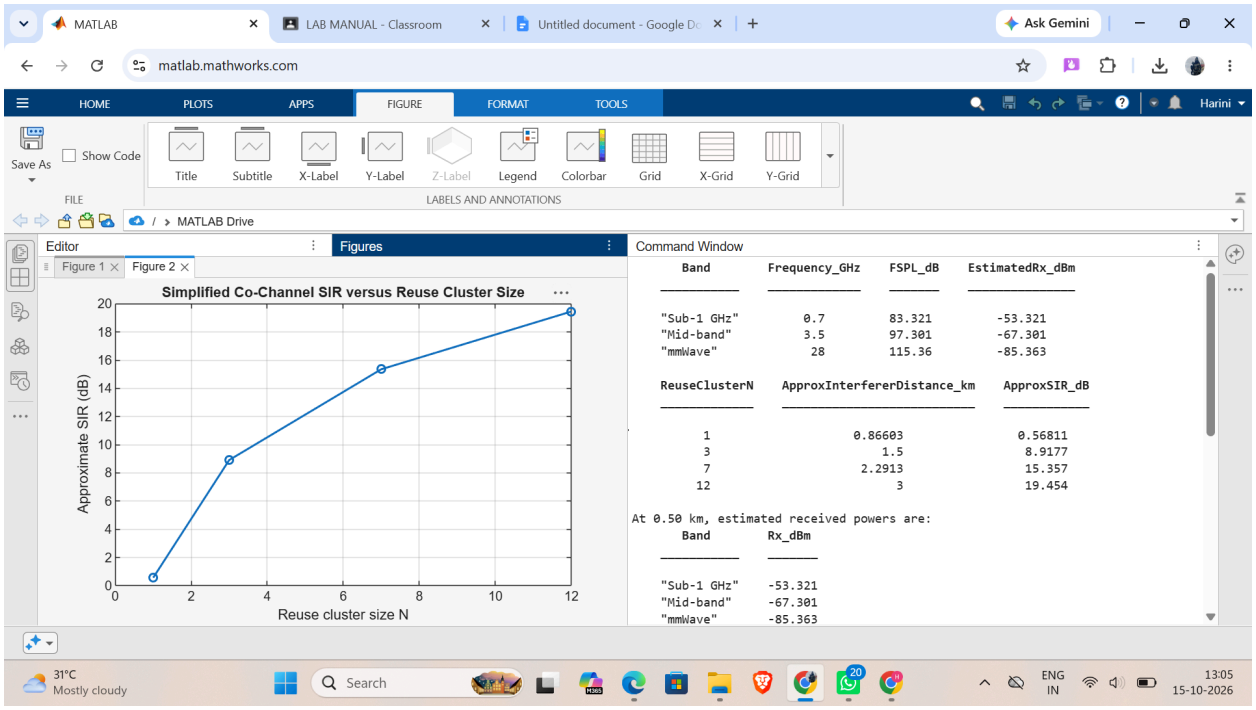
CODE:

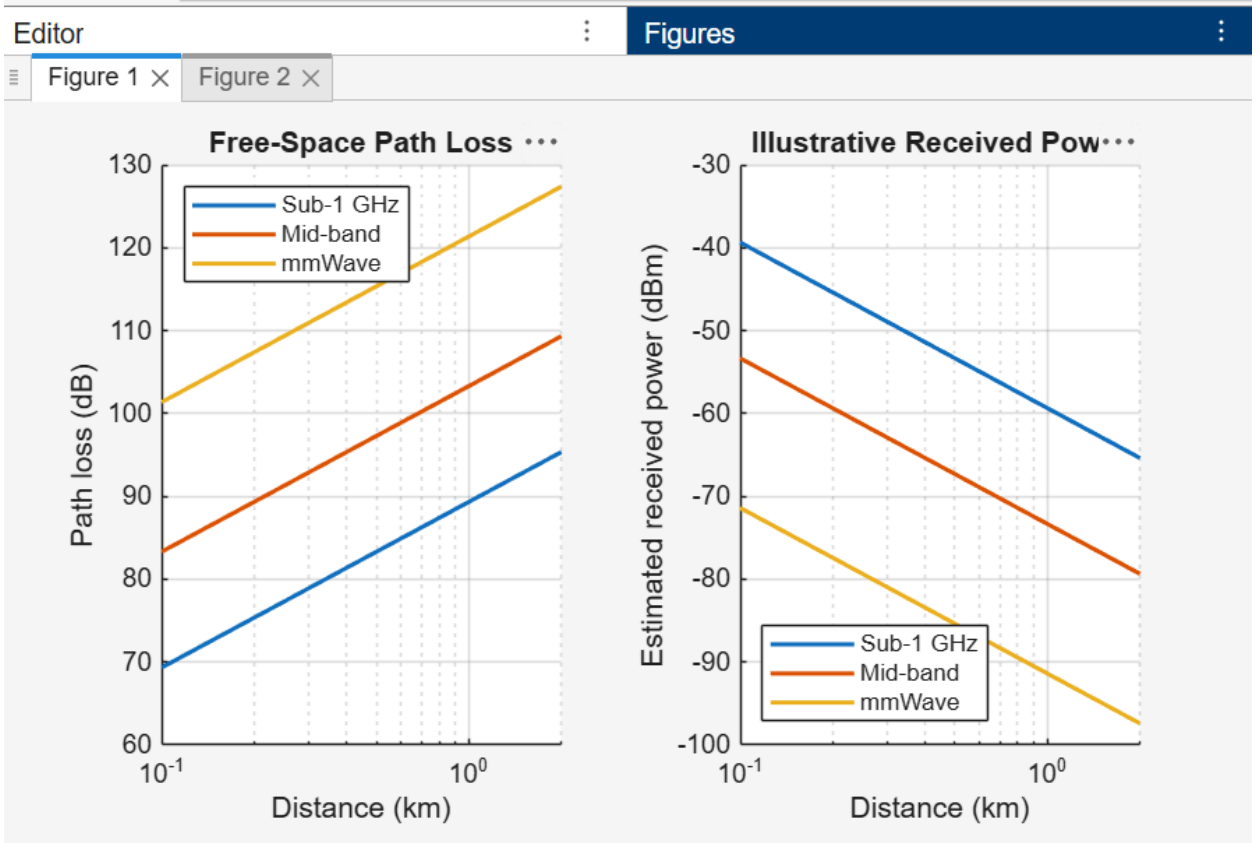
```
% Experiment 3: illustrative propagation and frequency reuse
clear; clc; close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
% Free-space path loss (d in km, f in MHz)
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
```

```
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');

legend(bandName,'Location','southwest'); title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

OUTPUT:


MATLAB
LAB MANUAL - Classroom
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Subtitle
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Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Editor
Figures
Figure 1
Figure 2
Simplified Co-Channel SIR versus Reuse Cluster Size
Approximate SIR (dB)
Reuse cluster size N
Command Window
Band Frequency_GHz FSPL_dB EstimatedRx_dBm
"Sub-1 GHz" 0.7 83.321 -53.321
"Mid-band" 3.5 97.301 -67.301
"mmWave" 28 115.36 -85.363
ReuseClusterN ApproxInterfererDistance_km ApproxSIR_dB
1 0.86603 0.56811
3 1.5 8.9177
7 2.2913 15.357
12 3 19.454
At 0.50 km, estimated received powers are:
Band Rx_dBm
"Sub-1 GHz" -53.321
"Mid-band" -67.301
"mmWave" -85.363
31°C
Mostly cloudy
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Editor
Figures
Figure 1
Figure 2
Free-Space Path Loss
Path loss (dB)
Distance (km)
Sub-1 GHz
Mid-band
mmWave
Illustrative Received Pow
Estimated received power (dBm)
Distance (km)
Sub-1 GHz
Mid-band
mmWave